

MIDTERM HOMEWORK A.Y. 2025/2026

As explained during the first introductory lecture of the course, the midterm homework is compulsory and can score up to **either 4 or 5 points maximum**. The final test is also compulsory, of course, and can score up to **either 25 or 26 points maximum**. Therefore, the final score will be calculated as follows:

Final score = midterm homework score + final test score

The groups will have to present their midterm homework on **Thursday, May 28, 2026**:

- from 2:30 p.m. to 4:30 p.m. in room A.0.6, Coppito 1, Edificio Ricamo, and
- from 4:30 p.m. to 6:30 p.m. in the usual room A1.3, Blocco 0.

This year the homework will be about Data as a Service and Ethics as a Service in a Service-Oriented Application

The following description is not intended to impose a rigid structure or too many constraints. Its purpose is to guide students toward the aspects they should consider while developing the project. Students are not required to address every single point, but their willingness and ability to cover as many relevant aspects as possible will certainly be taken into account in the evaluation.

The goal of this homework is to design and implement a small service-oriented application that integrates two concepts discussed in class:

- **Data as a Service**, that is, exposing data through reusable REST services.
- **Ethics as a Service**, that is, a separate service that evaluates requests, decisions, or operations according to explicit policies, producing a justified and traceable decision.

Students may freely choose the application domain: cinema, healthcare, mobility, environment, university services, tourism, employment, public administration, sport, culture, e-commerce, etc. The chosen domain should be rich enough to support both meaningful data queries and a non-trivial ethical discussion.

Project Description

Each group must implement an application composed of at least:

- A DaaS service exposing data through REST APIs.
- An EaaS service or module evaluating a request, decision, or operation of the application.
- A simple client demonstrating the interaction between user, DaaS, and EaaS.
- A final classroom presentation with slides and a working demo.

The DaaS part must build on the ideas of the examples shown in class: RDF datasets, SPARQL queries, REST endpoints, JSON output, and separation between data and application logic.

The EaaS part must build on the following concepts: structured requests, external policies, risk assessment, final decision, explanation, provenance, and audit trail.

Minimum DaaS Requirements

- The Data as a Service component must:
- use an RDF dataset created or adapted by the group;
- expose at least 5 meaningful REST endpoints;
- use SPARQL queries to query the dataset;
- return results in JSON format;
- include at least one query combining multiple conditions or relationships;
- document the dataset, main classes/relations, and available endpoints.
- Example endpoints:
- /items/{id}
- /items/category/{category}
- /items/location/{location}
- /items/year/{year}
- /items/search?...
- /items/risky
- /items/recommended

Minimum EaaS Requirements

The Ethics as a Service component must:

- receive a structured request from the client or backend;
- not trust a risk value directly declared by the caller;
- load external policies, for example JSON files;
- produce a structured evaluation including, e.g.:
 - o risk level;
 - o decision: PROCEED, REVISE, ESCALATE, or REJECT;
 - o rationale;
 - o applied policies;
 - o required actions, if any;
 - o reference to an audit record;
 - o store or expose an audit trace;
 - o distinguish between case analysis and governance decision.
 - o Policies must be coherent with the selected domain, for example:

- personal data privacy;
- fairness across user groups;
- security;
- transparency;
- data quality or provenance;
- responsible use of recommendations;
- protection of vulnerable users;
- licensing or source provenance.

Integration Between DaaS and EaaS

The application must show a case where data provided by the DaaS component is used to make, suggest, or support a decision, and that decision is evaluated by the EaaS component.

Examples:

- an app recommends public services, while EaaS checks fairness and transparency;
- a healthcare app displays data or suggestions, while EaaS checks privacy and risk;
- a tourism app recommends places, while EaaS evaluates bias, accessibility, or sustainability;
- a university app suggests courses or priorities, while EaaS evaluates equity and explainability;
- a recruiting app uses candidate data, while EaaS evaluates discriminatory proxies and accountability.

Each group must submit

- Project source code.
- RDF dataset used by the DaaS.
- Short documentation of the REST endpoints.
- At least 3 EaaS policies in an external format, such as JSON.
- At least 2 example requests:
 - one leading to PROCEED or REVISE;
 - one leading to ESCALATE or REJECT.
 - Audit records or examples of traces produced by the system.
 - Slides for the final presentation.
 - A short README with build, execution, and demo instructions.

Class Presentation

The presentation should last about 10-12 minutes for each member of the group, since larger groups are expected to develop more complex and more complete projects. For example, a group of two students should prepare a presentation of about 20-24 minutes, while a group of three students should prepare a presentation of about 30-36 minutes.

The presentation should include:

- selected domain and motivation;
- overall application architecture;
- RDF dataset description and main SPARQL queries;
- DaaS REST APIs description;
- EaaS workflow;
- policies defined by the group;
- example request evaluated by EaaS;
- audit/provenance of the decision;
- live demo;
- short critical reflection: what are the limits of your system? What should not be decided automatically?

Homework Assessment Criteria

IMPORTANT: Each group member is expected to have actively contributed to the homework and to be aware of the overall work carried out by the group. Therefore, the evaluation will take into account each student's individual performance and understanding of the work, not only the final group outcome.

Evaluation Criteria

- Technical correctness of the DaaS
- "Quality" of the REST/RDF/SPARQL integration
- "Quality" of the EaaS component
- Clarity of policies, decision, and audit
- Originality and coherence of the selected domain
- Quality of presentation and demo

Key Message for Students

You are not expected merely to “make two services work”. You must demonstrate that you are able to elaborate on the concepts underlying the project and that you understand that:

- data can be offered as a reusable service through APIs;
- a data-based decision is not automatically neutral;
- ethics must not be a final checkbox;
- policies, context, explanation, responsibility, and audit must become part of the software architecture.